

Week 01: Functions and types

Functional Programming in Lean

Learning goals

- ▶ Function definitions and type signatures
- ▶ `#eval` and `#check`
- ▶ Pure expressions

Core concepts

- ▶ Input types and result type
- ▶ Expression evaluation without side effects

Lean example

```
def double (n : Nat) : Nat := n + n  
#eval double 7  
#check double
```

Lean details

- ▶ `#eval`: evaluate an expression
- ▶ `#check`: display an expression's type
- ▶ Type errors: expected type and actual type

Live coding

- ▶ `double`: addition on `Nat`
- ▶ `larger`: conditional expression
- ▶ Predicted results before `#eval`

Check

Inferred type of `fun n : Nat => n + 1`

Exercise

- ▶ `clamp` on `Nat`
- ▶ Type error diagnosis